Mini-project (Big Data - CO5135) - Group 261_05

Scalable Real-Time Anomaly Detection on Operational Time-Series Data Using Apache Spark Structured Streaming

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Data Sources

Numenta Anomaly Benchmark (NAB)

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Labeled time series

- Timestamp - value CSV files
- Real and synthetic data
- Ground-truth anomaly windows

Selected Subset

realAWSCloudwatch

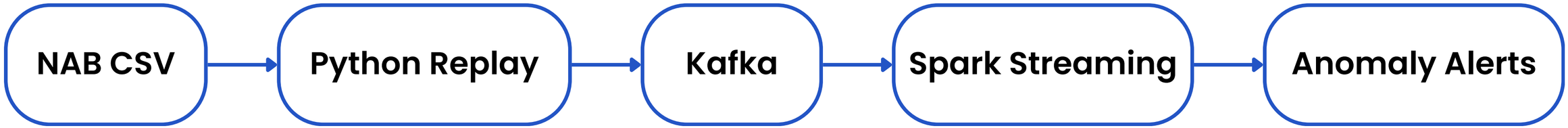
- AWS CloudWatch operational metrics
- Including CPU utilization



Illustrative time-series signal (not actual NAB values)

Source: github.com/numenta/NAB
Historical observations replayed to simulate a live stream.

Proposed Streaming Pipeline



Problem Statement

How can we achieve accurate and low-latency anomaly detection while efficiently processing continuous operational data streams?

Dynamic Data Patterns

Fixed detection baselines may become unreliable as normal behavior changes.

Detection Reliability

Balance anomaly detection performance against false alarms.

Processing Latency

Generate timely alerts while continuously consuming new records.

Streaming Scalability

Maintain stable processing performance under increasing input rates.

Proposed Approach

Implement a Kafka–Spark Structured Streaming pipeline with Static and Rolling Z-score anomaly detection.

Evaluation Objectives

- **Detection Quality:** NAB score and false-alarm rate
- **Streaming Performance:** Throughput and alert latency
- **Scalability:** Performance across input rates and Spark configurations